

CATHOLIC JUNIOR COLLEGE
JC2 Preliminary Examinations
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

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INDEX
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BIOLOGY

9648/03

Applications Paper and Planning Question

25 August 2015

Paper 3

2 hours

Additional Materials: Answer Booklet/Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
Total	

This document consists of **20** printed pages.

[Turn over

Answer **all** questions.

- 1 The artificial plasmid, pBR322, was constructed to act as a vector. It has often been used to insert human genes, such as the human insulin gene, into the bacterium, *Escherichia coli*.

The plasmid was constructed to include two genes, each giving resistance to a different antibiotic: an ampicillin resistance gene and a tetracycline resistance gene. The plasmid also has a target site for the restriction enzyme, *Bam*HI, in the middle of the tetracycline resistance gene.

A pBR322 plasmid was cut using *Bam*HI and the cDNA gene for human insulin inserted into it. Fig. 1.1 shows pBR322 and the recombinant plasmid.

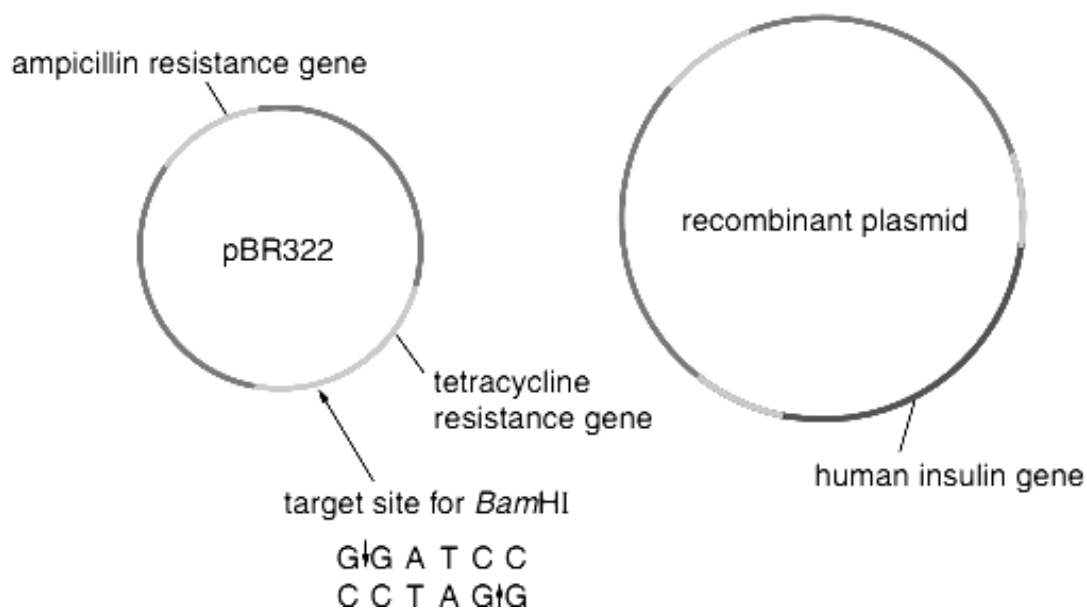


Fig. 1.1

- (a) With reference to Fig. 1.1, describe how a cDNA human insulin gene can be inserted into pBR322 that has been cut by *Bam*HI. [4]

ref. sticky ends ;
GATC and CTAG ;
complementary bases (pairing) ;
A to T and C to G ;
H-bonds (to sticky ends of plasmid) ;
(gaps in) sugar-phosphate backbones sealed by (DNA) ligase ;
AVP ; e.g. formation of phosphodiester bonds / ref. terminal transferase ;

[4 marks max]

- (b) Bacteria were then mixed with the recombinant plasmids. Those bacteria which had successfully taken up recombinant plasmids were identified using the following steps:
- step 1 – the bacteria were spread onto culture plates containing nutrient agar and ampicillin and incubated to allow colonies to form
 - step 2 – some bacteria from each of the colonies growing on these plates were transferred to plates containing nutrient agar and tetracycline, as shown in Fig. 1.2.

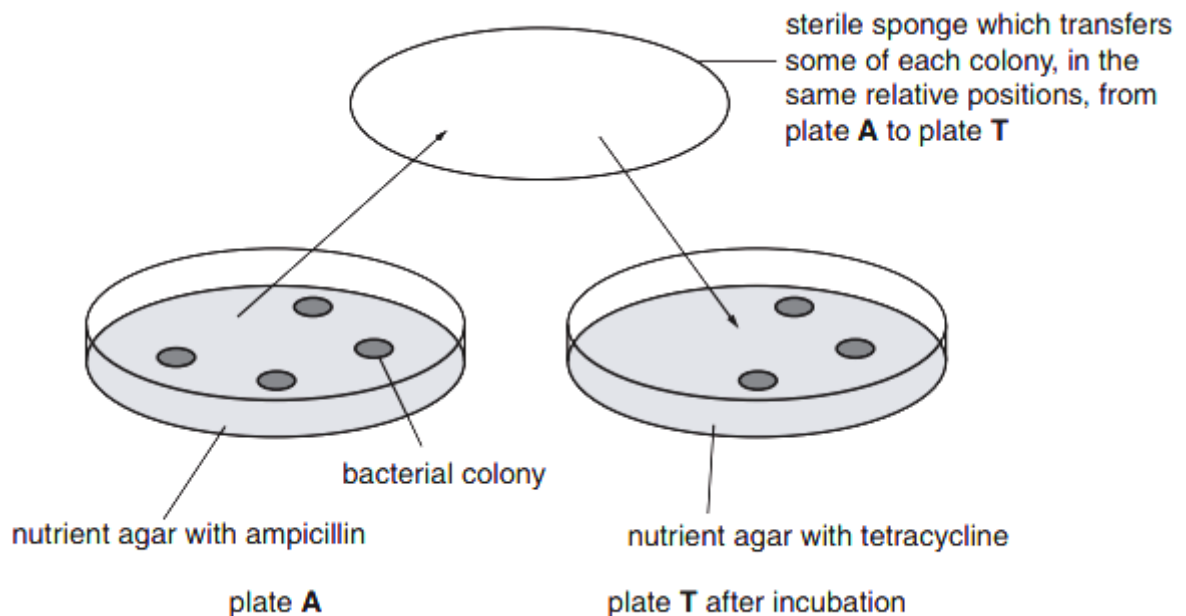


Fig. 1.2

- (i) Explain why the bacteria were first spread onto plates containing ampicillin. [3]

idea of identifying bacteria that, are transformed / have taken up plasmid / have taken up ampicillin resistance gene ;
 these bacteria have survived ;
 these bacteria may contain pBR322 or recombinant plasmid / plasmids taken up may not contain human insulin gene ;
 other bacteria have been killed ; [3 max]

- (ii) Explain why it is important, for identifying bacteria that have successfully taken up the recombinant plasmid, that on pBR322 the target site for *Bam*HI is in the middle of the tetracycline resistance gene. [3]

(*Bam*HI) breaks the tetracycline resistance gene ;
 (inserting human insulin gene) makes tetracycline resistance gene inactive ;
 colonies that are ampicillin-resistant but not tetracycline-resistant have taken up recombinant plasmid / insulin gene ; 4 colonies that survive on, tetracycline / both ampicillin and tetracycline / plate T, have not taken up the recombinant plasmid / insulin gene ; [3 max]

- (iii) Use a label line and the letter **C** to identify, on Fig. 1.2, a colony of bacteria that contain the recombinant plasmid.

Put your answer onto Fig. 1.2 on page 4.

[1]

Answer on Fig. 2.2 left hand colony on plate A;

- (c) Plasmid vectors carrying antibiotic resistance genes are now rarely used in gene technology.

- (i) Suggest why antibiotic resistance genes are now rarely used.

[2]

plasmids (easily) transferred between bacteria ;
 (bacteria of), same species / different species ;
 bacteria can acquire antibiotic resistance / renders antibiotic useless / AW ; [2 max]

- (ii) Suggest one type of gene that has replaced antibiotic resistance genes in plasmid vectors and indicate how its presence can be detected.

[2]

mark for gene and mark for how product detected 1 gene for β galactosidase ;
 blue colour from X-gal medium ;
 or gene for β glucuronidase (GUS) ;
 produces product that is easily stained blue ;
 or gene for, GFP / other fluorescent product ; R fluorescent / fluorescence, gene
 fluorescence detected when present ;
 or other gene ;
 how detected ;

[2 max]

[Total: 15]

2 Stem cells: recent progress in medical application.

In 2008, a woman had her damaged left bronchus replaced by one constructed using her own stem cells. A section of the trachea from a donor was treated in order to remove the cells of the donor, leaving only a framework of collagen. Adult stem cells derived from the woman's own bone marrow were then added to this collagen framework. A month later the transplanted tissue had developed its own blood supply. This was claimed to be the first successful transplant using tissues derived from the patient's own stem cells.

(a) Outline the advantages of using a patient's own stem cells to treat a damaged bronchus, as described in the passage, rather than transplanting a complete bronchus. [3]

- There will not be any immune response / tissue rejection since no foreign antigens, foreign tissue from the donated bronchus ;
- No need for immunosuppressant drugs ;
- Bronchus cells can be obtained / cultured in large quantities from the patient's stem cells ;
- Healthy culture of patient's stem cells can be stored for future needs ;
- Overcomes the problem of finding and locating suitable donors ;
- No need to do tissue / blood / histocompatibility matching ;
- Fewer, ethical issues than when using embryonic stem cells ;
- No risk of infection arising from donor tissue / no need to decide who is eligible ;

[max 3 marks]

(b) Adult stem cells derived from the patient's own bone marrow are described as multipotent, rather than totipotent.

Distinguish between multipotent cells and totipotent cells.

[1]

Multipotent cells give rise to limited range of cell types unlike totipotent cells which can give rise to, whole range of cells / whole organism ;

- (c) Once most stem cells differentiate, they lose their ability to proliferate indefinitely and differentiate into other types of cells. However, some fully differentiated cells can be stimulated to change back into stem cells in tissue culture. Such cells are called induced pluripotent stem cells (iPS cells). iPS cells regain the unique properties of stem cells.

In experiments with mice it was discovered that the introduction of four genes would cause certain fully differentiated cells to change to iPS cells.

There is evidence to suggest that the introduction of the four genes has caused an increase in the production of telomerase reverse transcriptase (TERT) in the fully differentiated cells.

- (i) State how the addition of the four genes caused an increase in the production of TERT. [1]
 They code for transcription factors (e.g. activators);
 They code for chromatin remodeling proteins (e.g. Histone deacetylases);
 AVP ; [max 1 mark]
- (ii) Explain how TERT may help to change the cells back into stem cells. [2]
 • TERT plays a part in increasing the length of the telomere; (mechanism involving) TERT extends chromosome ;
 • In doing so counteracts / buffers end-replication problem to prolong / to allow cell division / DNA replication to continue over a longer period of time, this gives the iPS cell capacity to divide repeatedly ;

A cybrid (cytoplasmic hybrid cell) is produced as shown in Fig. 2.1.

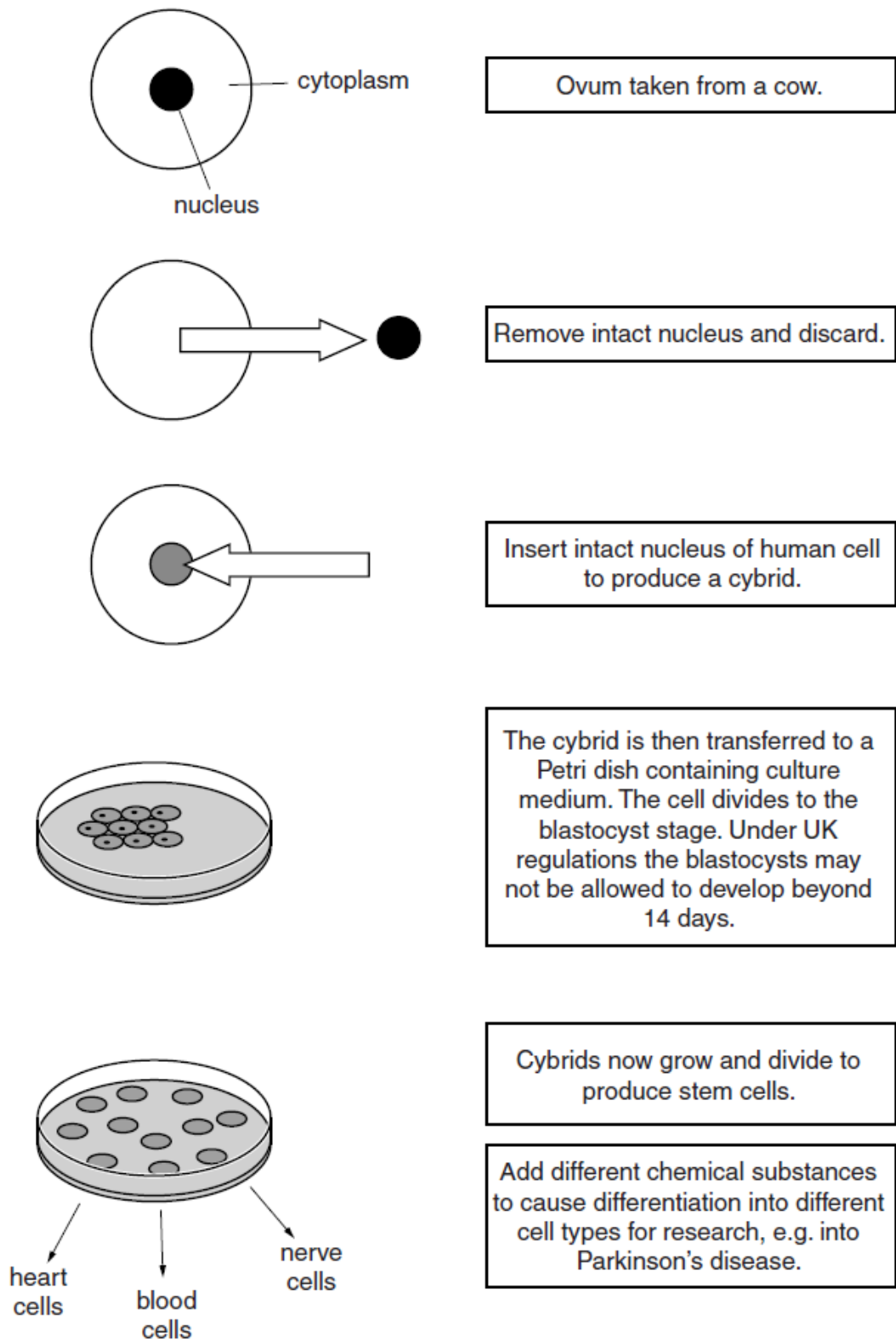


Fig. 2.1

- (d) The DNA of such a cybrid is 99.6% human. The remaining 0.4% of the DNA is in the cytoplasm. Explain the presence of the 0.4% DNA in the cytoplasm. [2]
 in the mitochondria ;
 (mitochondrial DNA) from the cow / not of human origin ;

- (e) Some people argued that it is unethical to allow the production of cybrids.

State whether you agree **or** disagree that this is unethical **and** explain why you reached this decision. [2]

unethical:

- presence of human and animal DNA means it is partly human and partly cow / a human animal hybrid which would not happen in nature;
- claims of the benefits of embryonic stem cell research are over-rated / few (if any) examples of success in medical applications ;
- dangerous precedent / thin end of the wedge argument / current applications (may be benign) but will lead to abuse in future ;
- possibility of unforeseen (irreversible) consequences ;
- unnecessary because, there already are / will soon be, alternative techniques (to the use of embryonic stem cells) e.g. use of adult stem cells/umbilical cord cells ;

ethical:

- the amount of non-human DNA is very small/negligible ;
- protocol limits keeping 'embryo' to 14 days (to guard against abuse) so cannot develop into a whole organism ;
- provides a more ethical alternative to the use of embryonic stem cells from humans
- can provide many more stem cells than is possible from embryonic stem cells ;
- it would be unethical not to use cybrids to relieve, human suffering / e.g. diseases such as Parkinson's disease ;
- techniques which offer alternatives to embryonic stem cells builds on work done initially on embryonic stem cells ;
- idea that ovum has bovine proteins which would be rejected at implantation ;
- Allow other reasonable arguments on either side ; [max 2 marks]

If a candidate does not express an opinion, limit score to one mark unless an attempt has been made to argue that it is a finely balanced matter.

Ignore references to 'playing God' and 'against religion'.

[Total: 11]

- 3 (a) To produce *Bt* maize, the procedure to genetically modify plant cells, shown in Fig.3.1, was carried out.

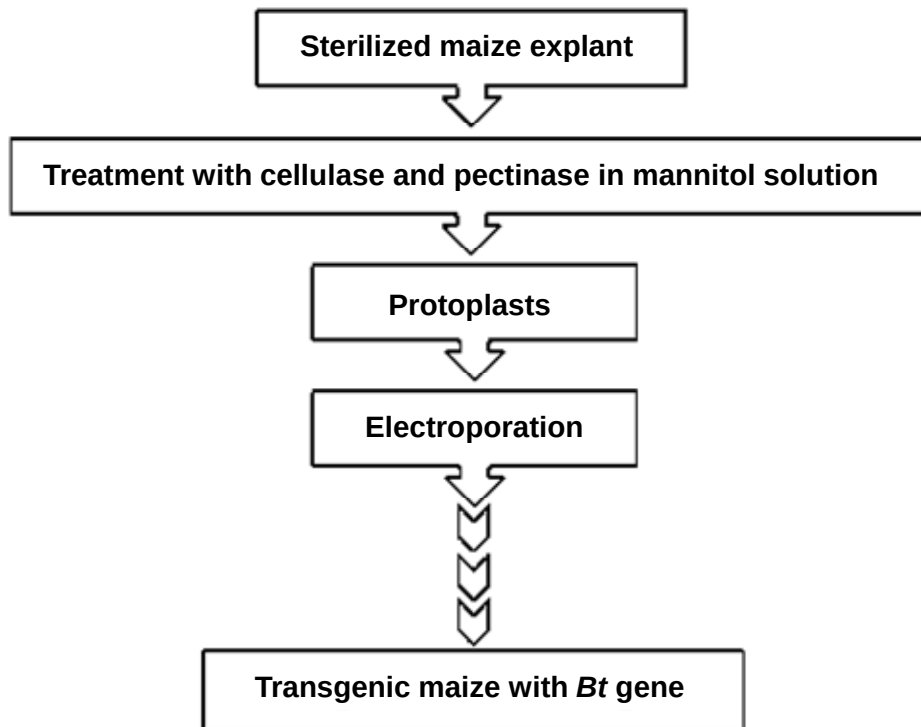


Fig. 3.1

- (i) Name a vector suitable for genetically modifying plant cells that can be used in this procedure. [1]
Agrobacterium tumefaciens / Ti plasmid / Gene Gun ;
- (ii) Explain why the sterilized explant was treated with cellulase and pectinase before electroporation. [2]
- To remove cell wall to form protoplasts;
 - Protoplasts which are lacking in cell wall can then take up plasmid containing *Bt* gene during electroporation more readily / cell wall is a barrier to interfere with the uptake of *Bt* gene during electroporation;

Addition of mannitol to water lowers the water potential of water. In the preparation of protoplasts, mannitol must be present. In the absence of mannitol, no protoplasts are formed even if the sterilised explant was treated with cellulase and pectinase.

- (ii) State what happens to the protoplasts if mannitol was not used. [1]
 The protoplasts undergo lysis / will burst due to intake of water ;

Fig. 3.2 shows the proportion of cotton, maize and soybeans grown in the USA that are genetically modified in two different ways:

- HT crops are modified to be resistant to the herbicide glyphosate.
- Bt crops are modified to express the Bt toxin which kills insect pests.

percentage of total
crop grown that is
genetically modified

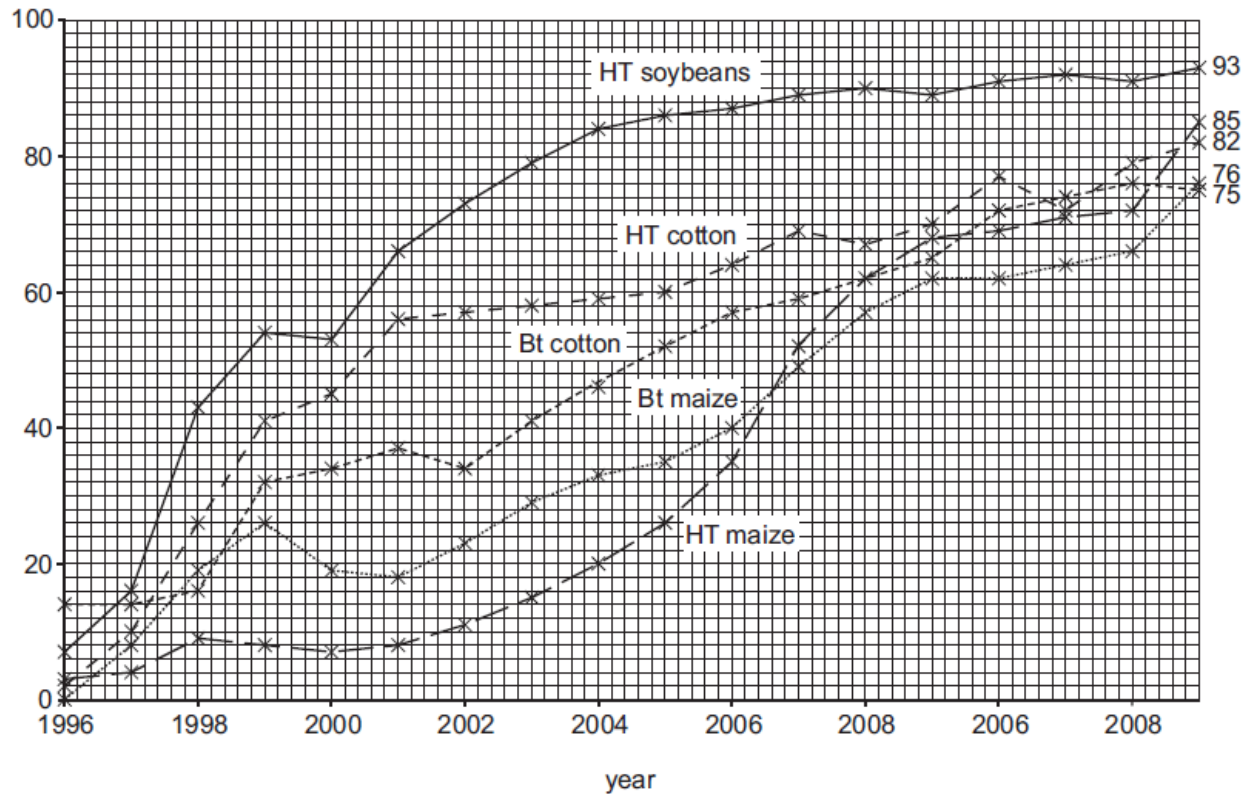


Fig 3.2

- (b) With reference to Fig. 3.2, describe the change in the percentage of genetically modified cotton grown in the USA since 1996. [2]

General description of the trend:

1. % of genetically modified Bt cotton grown increased from 16% in 1996 to 75% in 2013, by a magnitude of 59% over 17 years
2. % of genetically modified HT cotton grown increased from 2% in 1996 to 82% in 2013 by a magnitude of 80% over 17 years

Comparison of increase in % of total crop grown:

3. Increase in % of total crop grown for HT cotton over the 17 years, is greater compared to that for modified Bt cotton - 80% vs 59%
4. For Bt cotton or HT cotton, the steepest / fastest increase in % is observed between 1996 and 1999; [max 2 marks]

- (c) The HT crops received two new genes that gave resistance to glyphosate herbicide and also a marker gene called GUS. The parts of the plant that express the GUS gene turn blue when dipped into a colourless chemical substrate.

Explain why the GUS gene was also transferred to the genetically modified crops. [2]

- To check whether the gene transfer is successful as only the transgenic crop plant will have parts (with the GUS gene) turning blue when dipped in the colourless chemical substrate ;
- To detect and determine which parts of the plant are able to express new transgenes / introduced genes, as these parts will turn blue when dipped into a colourless chemical substrate ;
- Ease of use in using GUS gene marker, to track / visualise the location of the gene (compared to antibiotic resistance markers) ; [max 2 marks]

- (d) Fig. 3.3 shows the increase in the number of weed species resistant to glyphosate herbicide and triazine herbicides since 1970.

Crops have not been genetically modified to resist triazine herbicides.

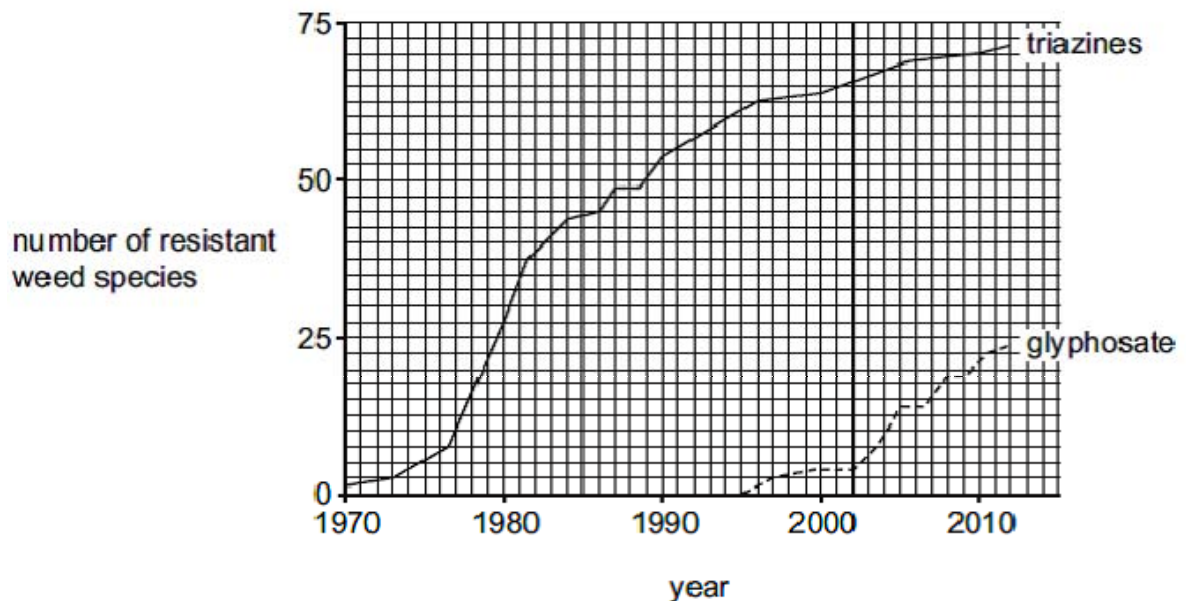


Fig 3.3

- (i) With reference to Fig. 3.2 and Fig. 3.3, justify the statement that the rise in glyphosate-resistant weeds has resulted from the introduction of genetically modified crops. [2]
- Number of glyphosate-resistant weed species only increased after 1995 / 1996 ;
 - This was when, genetically modified crops resistant to herbicide / HT crops, were introduced ;
- (ii) Give **one** piece of evidence from Fig. 3.3 to support the idea that the development of weeds resistant to herbicides is **not** due to horizontal gene transfer from genetically modified crops. [1]
- Idea that no triazine-resistance genes existed in crops / No genetically modified crops were grown to be triazine resistant were grown prior to 1970s when weeds developed triazine resistance ;
 - Idea that weeds developed triazine resistance prior to gene technology / genetic modification;
- (iii) Outline how herbicide resistance in weeds arises and spreads without horizontal gene transfer. [3]
- Herbicide resistance / variation in herbicide resistance in the weed population arise due to spontaneous / random, genetic mutation
 - Weeds without herbicide resistance / herbicide resistance allele will be destroyed, in the presence of herbicide spraying (as selection pressure)
 - Those with herbicide resistance (allele) will be selected for / conferred selective advantage; these will survive to reproductive age, breed / reproduce successfully to pass on, herbicide resistance allele to subsequent generations
 - Over time, frequency of the herbicide resistance allele increases; [max 4 marks]

[Total: 14]

4 Planning question

The enzyme urease is a catalyst for the hydrolysis of urea in solution, for example in the breakdown of urea in soils by microorganisms, forming ammonia and carbon dioxide.

As the reaction proceeds, the ammonia released dissolves, causing the pH to increase.

You are required to plan an investigation to compare the activity of urease free in solution and urease immobilised in alginate beads.

You are provided with the following equipment which you may or may not use in your plan. However, you may **not** use any additional equipment in your plan.

- calcium alginate beads all of uniform size prepared with a 50gdm^{-3} urease solution (you may call this immobilised urease)
- 50gdm^{-3} urease solution (you may call this free urease)
- 1.0mol dm^{-3} urea solution
- distilled water
- beakers and flasks of different sizes
- stop watch or electronic timer
- pH probes and meters
- thermometer
- thermostatically-controlled water baths
- graduated pipettes and pipette fillers
- filter funnels
- syringes
- glass rods for stirring
- test-tubes and boiling tubes
- test-tube racks

Your plan should

- include a clear statement of the hypothesis or prediction
- identify the key variables
- give full details and explanations of the procedures that you would adopt to ensure that the results are as precise and reliable as possible
- show how you would present and analyse your results
- include a brief risk assessment
- be written in clear scientific language.

You may include a diagram or diagrams in your plan.

[Total: 12]

[L3]

9790/03 Cambridge Pre-U 2012 Paper 4/3

Suggested Mark Scheme:

Planning

P = defining the problem

M = methods

Analysis, conclusions and evaluation

D = interpretation of data or observations and identifying sources of error

C = drawing conclusions

E = suggesting improvements and evaluation

Suggested Mark Scheme:

Sections	Learning outcomes	Expected answer	Mark
P defining the problem	- identify the dependent and independent variables in the experiment or investigation - express the aim in terms of a prediction or hypothesis, and express this in words and in the form of a predicted graph - identify the variables that are to be controlled	1 Hypothesis or prediction ; e.g. rate of hydrolysis is faster using free enzyme/quantity of urea hydrolysed over time is greater with free enzyme/immobilised urease catalyses reaction over much longer period of time 2 Theory to support candidate's hypothesis or prediction ; e.g. refs to accessible active sites/diffusion of substrate into alginate beads/stability of enzyme in alginate beads 3 Outline of strategy I e.g. method of following the reaction taking samples at intervals and calculating the initial rate 4 Justification/evaluation, of strategy ; e.g. can only alter concentration of immobilised enzyme by changing number of beads/limitations of colour comparison these could be awarded at the end of the plan 5 Method of determining, pH / (the concentration) of ammonium carbonate, at intervals ; e.g. use of pH indicator, to follow colour change 6 At least two control variables ; e.g. temperature, concentration of urea solution, volumes used, number of beads 7 Risk assessment ; Ref to hazard <u>and</u> precaution <i>some points may be taken from a diagram or a flow or sequence diagram</i>	[max 6]

Sections	Learning outcomes	Expected answer		Mark
P methods	- describe the method to be used to vary the independent variable, and the means that they will propose to ensure that they have measured its values accurately - describe how the dependent variable is to be measured - describe how each of the other key variables is to be controlled - explain how any control experiments will be used to verify that it is the independent variable that is affecting the dependent variable and not some other factor - describe the arrangement of apparatus and the steps in the procedure to be followed - suggest appropriate volumes and concentrations of reagents and explain how different concentrations would be prepared - assess the risks of their proposed methods - describe precautions that should be taken to keep risks to a minimum - draw up tables for data that they might wish to record - describe how the data might be used in order to reach a conclusion	8	use range of concentrations of urea ;	
		9	use range of concentrations of urease ; <i>to find suitable concentrations to make comparison</i>	
		10	dilution table(s) included ; Accept: ratios	
		11	method to ensure concentration of urease in reaction mixtures is the same for both free and immobilised enzyme ;	
		12	urea solution mixed with pH indicator ;	
		13	equilibration in water bath ;	
		14	mixing, urease/beads, and urea solution at time = 0 ;	
		15	staggered start ;	
		16	samples taken at stated intervals ;	
		17	uncertainty/precision, of results ;	
		18	plot results and take gradient to give initial rate ;	
		19	colour standard set up at known pH ;	
		20	time taken (t) to reach colour standard	
		21	recorded ;	
		22	rate = 1/t; Accept: 1000/t, etc. colour change followed in colorimeter ;	
		23	repeats/replicates (calculate means) ;	
		24	calculate, standard deviation/standard error ;	
		25	ref to use of <i>t</i> -test to see if rates are significantly different ;	
		26	plot results on appropriate graph (bar or line) ;	
				[max 10]

5 Free-response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate,
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

(a) Explain the different uses of RFLP analysis.

[6]

Disease detection

- use of RFLP markers as a surrogate for genetic diseases for which the gene is unknown ;
- direct screening for disease-causing alleles ;
- ref. use before conception to screen parent's DNA / use after amniocentesis to screen fetal DNA ;

[max 2 marks]

DNA fingerprinting

- ref. to VNTR (variable number of tandem repeats) or STR (short tandem repeats) ;
- identify criminals / determine paternity / determine whether an applying immigrant is really a close relative of already established residents ;

[max 2 marks]

Genomic mapping

- use RFLP markers as genetic markers for a particular location/locus in the genome ;
- make genome wide linkage map / find out the order of RFLP markers on a chromosome / ref. frequency of crossing over between 2 RFLP markers / indicate the distance between two loci on a chromosome ;

[max 2 marks]

- (b) Describe the advantages and limitations of plant tissue culture as compared to traditional agricultural methods such as grafting and cutting. [8]

Advantages

- Tissue culture allows the rapid multiplication of plants as subculturing can be done, giving rise to large numbers of genetically identical plants ;
- Using meristematic regions of a plant for tissue culture eliminates the transmission of plant diseases caused by bacteria and viruses ;
- Micropropagation can be used for plant species that are difficult to propagate by traditional methods. For example, bananas and orchids which have very low seed germination rates ;
- Micropropagation is not limited by seasons/weather conditions so plant can be produced continuously to meet consumer demand for out-of-season plant/regardless of weather conditions, unlike the traditional methods of propagation which are often limited by seasonal variations ;
- Micropropagation is an ideal form in which to export plants as large numbers can be air-freighted easily and cheaply ;
- With micropropagation, it is possible to standardise the conditions for growth and obtain many batches of identical plants ;
- Plants grown using tissue culture can be easily subjected to genetic engineering to introduce new desired traits ;

[max 4 marks]

Disadvantages

- Plant tissue culture produces batches of plants that are genetically identical and therefore susceptible to the same bacteria and viruses. Contamination by bacteria, if it does occur, can spread very rapidly as the plant cells are genetically identical. This causes high losses in the production line in a short time ;
OR Because the plant batches are all genetically identical, they may be wiped out by a single selection pressure, incurring huge losses in a short time ;
- Micropropagation requires sophisticated laboratory facilities and sterile laboratory conditions and special nutrient media. It also requires trained personnel with specialised skills and a lot of effort to transfer the plantlets from the laboratory to the soil ;
- The cost of micropropagation is high due to the facilities needed and high labour input ;
- Genetically-off types may occur when tissue culture is used for cloning plants, resulting in less than ideal plant clones ;

[max 4 marks]

(c) Discuss the benefits and ethical concerns of the human genome project.

[6]

Benefits: State + describe one example within category

- Molecular Medicine:
 - Improved diagnosis of disease/
 - Earlier detection of genetic predispositions to disease/
 - Rational drug design/
 - Gene therapy and control systems for drugs/
 - Pharmacogenomics "custom drugs";
- DNA Forensics (Identification) ;
 - Identify potential suspects whose DNA may match evidence left at crime scenes;
 - Exonerate persons wrongly accused of crimes/
 - Identify crime and catastrophe victims/
 - Establish paternity and other family relationships/
 - Identify endangered and protected species as an aid to wildlife officials (could be used for prosecuting poachers)/
 - Detect bacteria and other organisms that may pollute air, water, soil, and food;
 - Match organ donors with recipients in transplant programs/
 - Determine pedigree for seed or livestock breeds/
 - Authenticate consumables such as caviar and wine ;
- Risk Assessment ;
 - Assess health damage and risks caused by radiation exposure, including low-dose exposures/
 - Assess health damage and risks caused by exposure to mutagenic chemicals and cancer-causing toxins/
 - Reduce the likelihood of heritable mutations ;
- Bioarchaeology, Anthropology, Evolution, and Human Migration ;
 - Study evolution through germline mutations in lineages/
 - Study migration of different population groups based on female genetic inheritance/
 - Study mutations on the Y chromosome to trace lineage and migration of males/
 - Compare breakpoints in the evolution of mutations with ages of populations and historical events ;
- Energy and Environmental Applications ;
 - Use microbial genomics research to create new energy sources (biofuels)/
 - Use microbial genomics research to develop environmental monitoring techniques to detect pollutants/
 - Use microbial genomics research for safe, efficient environmental remediation/
 - Use microbial genomics research for carbon sequestration ;
- Agriculture, Livestock Breeding, and Bioprocessing ;
 - Disease-, insect-, and drought-resistant crops;

Healthier, more productive, disease-resistant farm animals;
 More nutritious produce;
 Biopesticides;
 Edible vaccines incorporated into food products;
 New environmental cleanup uses for plants like tobacco ;

[max 3 marks]

Ethical Concerns

- Fairness in the use of genetic information by insurers, employers, courts, schools, adoption agencies, and the military, among others.
- Privacy and confidentiality of genetic information;
- Psychological impact and stigmatization due to an individual's genetic differences;
- Reproductive issues including adequate informed consent for complex and potentially controversial procedures / use of genetic information in reproductive decision making / reproductive rights;
- Clinical issues including the education of doctors and other health service providers, patients, and the general public in genetic capabilities, scientific limitations, and social risks / implementation of standards and quality-control measures in testing procedures;
- Uncertainties associated with gene tests for susceptibilities and complex conditions (e.g., heart disease) linked to multiple genes and gene-environment interactions;
- Conceptual and philosophical implications regarding human responsibility, free-will versus genetic determinism / concepts of health and disease;
- Health and environmental issues concerning genetically modified foods (GM) and microbes;
- Commercialization of products including property rights (patents, copyrights, and trade secrets) and accessibility of data and materials;

[max 3 marks]

[Total: 20]